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Machine Learning Techniques

Unit-4

* Artificial Neural Network +2021-22.
* Perceptron
* Gradient Descent
* Delta Rule
* Back propagation algorithm - 2020-21
* Deep learning
* CNN - 2022-23 and its layer
* 1D - 2D CNN
* Training of Network
* Case studies.

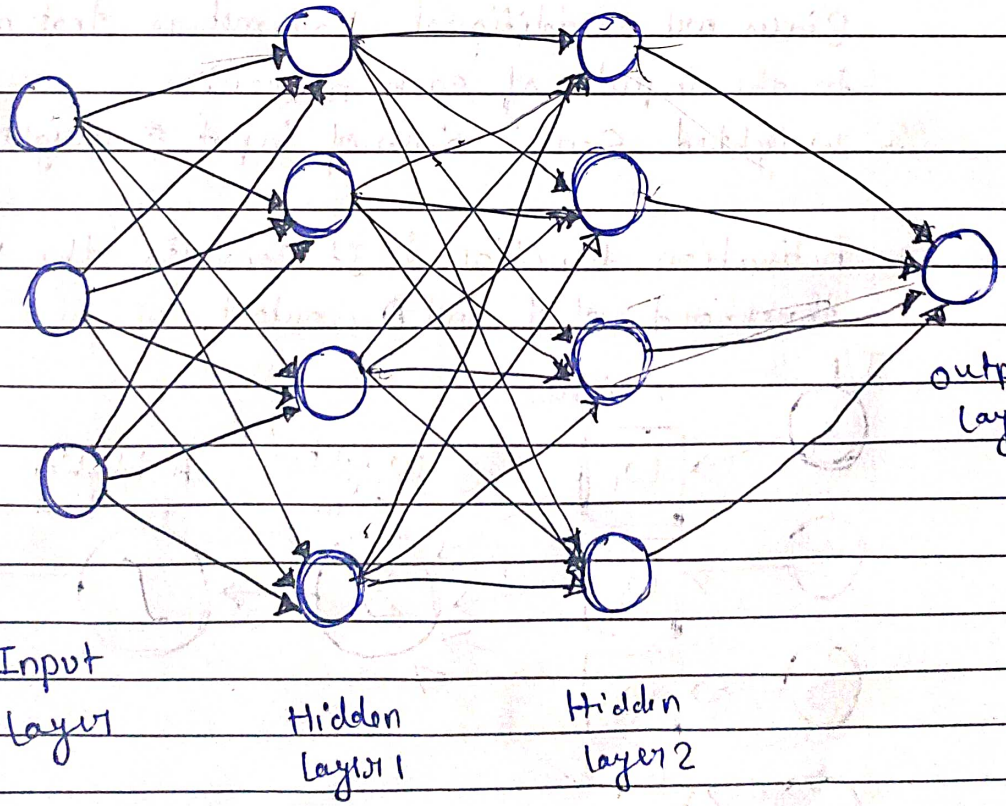
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* Artificial Neural Network (ANN)

- Inspired by structure and function of the human brain.
- Computational model that consists of interconnected nodes, called neurons, which are organized into layers.
- Each neuron receives input from other neurons, processes that input, and then sends an output signal to other neurons in the network.
- The input layer receives input data, which is then passed through one or more hidden layers, where data processed.
- The final output is produced by the output layer, which provides the result of the computation.



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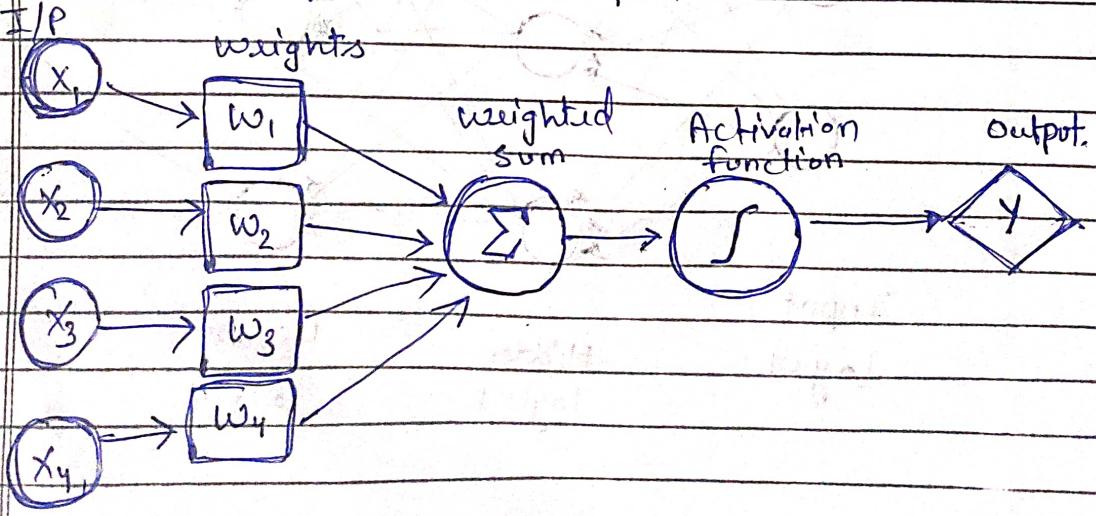
Application of ANN

- Image and speech recognition
- Natural language processing and
- Time-series prediction.

* Perceptron

→ It is the simplest form of a neural network.
→ also known as single layer neural network.
→ It contains 4 main parameters : input values, weights and Bias, net sum and an activation function.

1. Input Layer : Represent input features.
2. Weight and Bias : Strength of connection b/w neurons.
   Biases are additional parameters that are added to the input of each neuron.
3. Weighted Sum : Sum of input * weights (plus a bias term)
4. Activation function : It converts the values in threshold of 1 and 0 output.



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* Gradient Descent:

→ It is an optimization algorithm used to ~~maint~~ minimize the cost function in the process of training a machine learning model.

→ The goal is to minimize the difference b/w the predicted output and the actual output.

→ The optimization process involves adjusting these parameters iteratively:

- Initialize parameters ✓
- Calculate the Gradient ✓
- Update Parameters. ✓
- Repeat step 2 & 3.

* Delta Rule: ✓

→ It is a gradient descent learning rule.

→ This rule is used to update the weights in the direction that reduces the error between the predicted output and the target output.

$$\Delta w_{ij} = \eta \cdot (t - y) \cdot x_i$$

where,

- Δw_{ij} → change in weight b/w i and j.
- η → learning rate ✓
- t → target output ✓
- y → Actual output ✓
- x_i is the input from neuron on i. ✓

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[2020-21]

* Back propagation algorithm

→ It is an algorithm used in the training of fed forward neural networks for supervised learning.

→ This Algo. that is designed to test for errors working back from output nodes to input nodes.

→ Back propagation efficiently computes the gradient of the loss function with respect to the weights of the network.

→ updating weights to minimize loss.

→ Diagram same as ANN with backward arrows

* Self-Organizing Map (SOM) Algorithm

→ SOM provides a data visualization technique which helps to understand high dimensional data by reducing the dimensions of data to a map.

→ represents clustering

→ also known as Kohonen Map. (world map).

Variants ⇒ (GSOM, S-SOM, I-SOM)
(Batch Self-Organizing Map, Multi layer SOM.)

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* Deep learning

→ It is a subfield of machine learning that involves the use of ANN to model and solve complex problems.

→ "Deep" refers to the use of multiple layers in neural network. These networks, often called deep neural networks.

→ Used in
- image and speech recognition.
- Natural language processing etc.

* Concept of Convolutional neural network 22-23

→ A CNN is a specialized type of neural network designed for tasks involving grid-like data, such as images or sequences. [matrix]. [image pixel] [2d]

→ It is a specialized type of neural network model designed for working with two dimensional image data. Although they can be used with one-dimensional and three-dimensional data.

→ CNNs have proven to be highly effective in computer vision tasks, including [image classification, object detection, and image generation.]

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AKTU - 2022-23

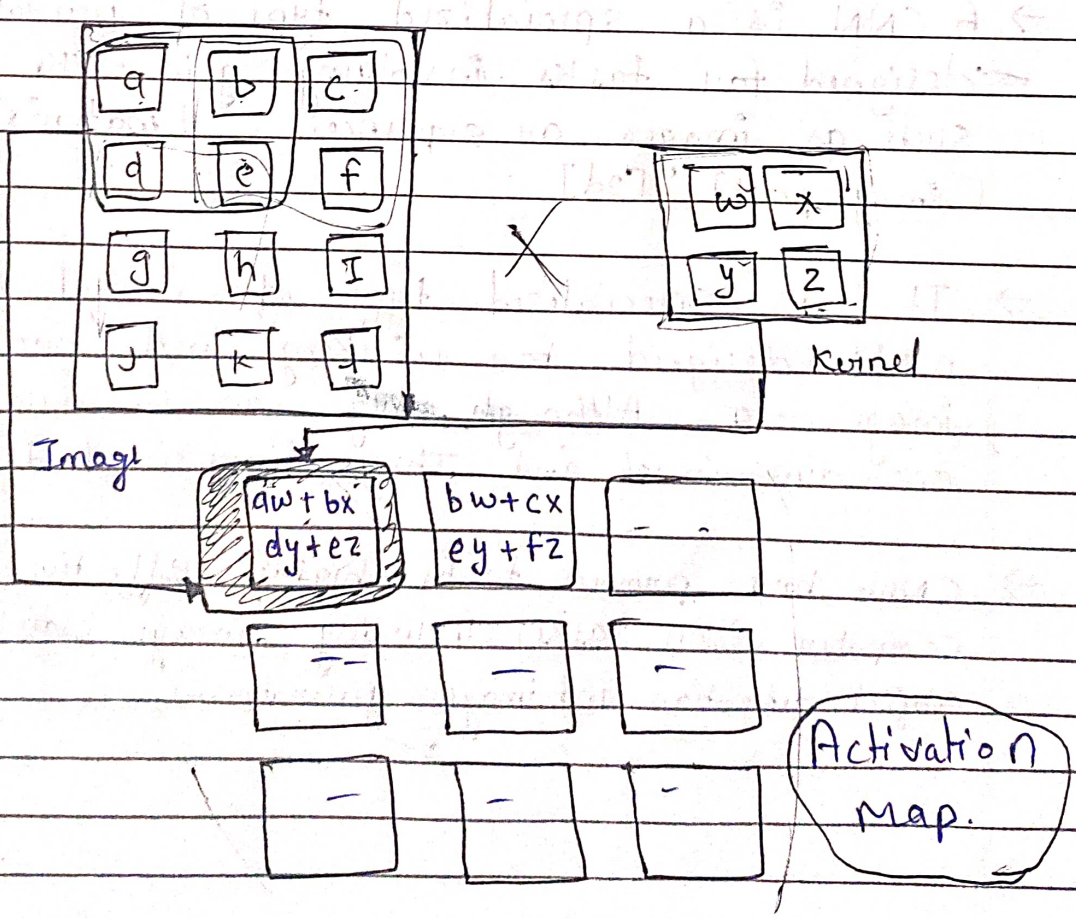
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Types of Layers:

① Convolutional Layers:

→ CNNs use convolutional layers that apply convolution operations to the input data.
→ A convolution involves sliding a small filter (also called as Kernel)
→ This operation captures local patterns, such as edges or textures.

• Multiple filters: Each convolutional layer typically consists of multiple filters, and each filter captures a different aspect or features of the input.



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② Activation function

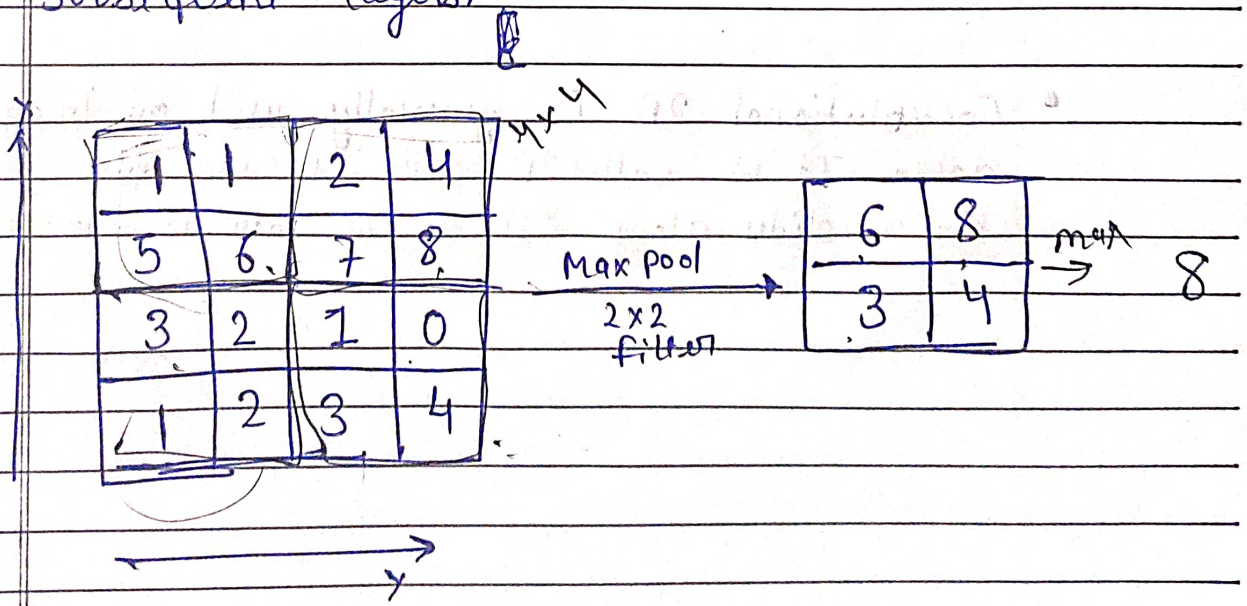
→ After the convolution operation, an activation function (commonly ReLU - Rectified Linear Unit) is applied element-wise to introduce non-linearity [This allows the network to learn more complex mappings]

③ Pooling Layers

→ It reduce the spatial dimensions of the input by down-sampling. [eg → 2D matrix to single value]

→ Common pooling operations (maximum value in a region) and (calculating the average).

• Subsampling : Pooling helps make the representation invariant to small translations and distortions in the input. It also reduces the computational load and the number of parameters in subsequent layers.



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④ Fully Connected Layers

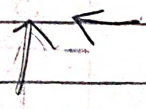
→ After several convolutional and pooling layers, one or more fully connected layers are often added.

→ These layers take the high-level, spatially invariant features learned by the convolutional layers and use them for classification or regression tasks.

* 1D & 2D Convolutional Neural Network

- A 1D CNN is primarily used for sequential data, such as time series or text data.
  → It allows the network to learn patterns and features along a sequence.

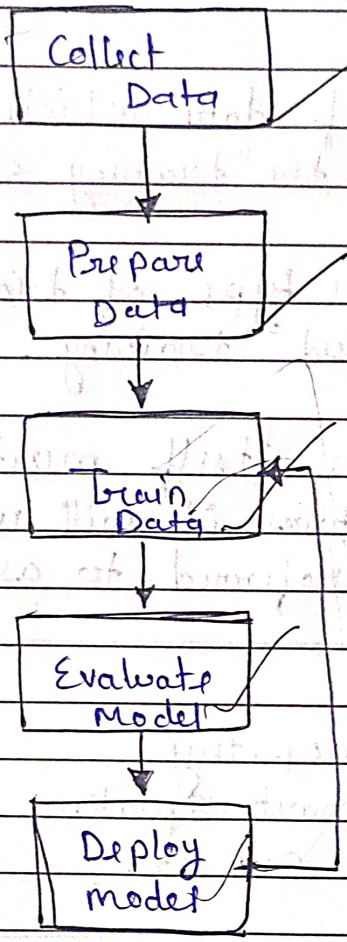
- Convolutional 2D is generally used on image data. It is called 2D CNN because the kernel slides along 2D on the data as shown.



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* Training of Network



1. → Once a network has been structured for a particular application, that network is ready to be trained.

2. → To start this process the initial weights are chosen randomly. Then, the training or learning begins.

3 → There are two approaches.

a. In Supervised training, both the input and the output provided.

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b. Errors are then propagated back through the System. This process occur over and over.

c. The set of data which enables the training is called the "training set".

d. The other type of training is called unsupervised training. - provide only input.

e. The system itself must then decide what features it will use to group the data. - referred to as Self-organization

Case study
* Diabetic Retinopathy
* Building a smart Speaker
* Self-deriving car etc.

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Road to 5K.

Describe diabetic retinopathy on the basis of deep learning.

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1. Diabetic Retinopathy (DR) is one of the major causes of blindness in the western world. Increasing life expectancy, indulgent lifestyles and other contributing factors mean the number of people with diabetes is projected to continue rising.

2.Regular screening of diabetic patients for DR has been shown to be a cost-effective and important aspect of their care.

3.The accuracy and timing of this care is of significant importance to both the cost and effectiveness of treatment.

4.If detected early enough, effective treatment of DR is available; making this a vital process.

5.Classification of DR involves the weighting of numerous features and the location of such features. This is highly time consuming for clinicians.

6.Computers are able to obtain much quicker classifications once trained, giving the ability to aid clinicians in real-time classification.

7.The efficacy of automated grading for DR has been an active area of research in computer imaging with encouraging conclusions.

8.Significant work has been done on detecting the features of DR using automated methods such a support vector machines and k-NN classifiers.

9.The majority of these classification techniques arc on two class classification for DR or no DR.

Using artificial neural network how we recognize speaker.

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• Objective: Develop a speaker recognition system using Artificial Neural Networks (ANNs).

• Data Collection: Collect diverse audio samples from various speakers, covering different speech patterns and recording Conditions.

• Feature Extraction: Extract relevant features, such as MFCCs, from the audio data to capture unique voice characteristics.

• Model Architecture: Design an ANN architecture with input, hidden, and output layers suitable for speaker recognition.

• Training and Tuning: Train the model using optimization algorithms and tune hyperparameters for optimal performance.

• Validation and Testing: Regularly evaluate the model on validation sets, ensuring it generalizes well to unseen speakers during testing.

• Deployment: Deploy the trained model for real-world speaker recognition tasks, integrating it into applications like security systems or voice assistants.

• Continuous Improvement: Monitor the model's performance in real-world scenarios, considering continuous learning strategies for adaptation to new speakers or changing conditions.

Artificial intelligence plays important role in self-drivingcar explain.

• Objective: Develop a self-driving car system using Artificial Neural Networks (ANNs) for autonomous navigation.

• Data Collection: Gather diverse sensor data, including images, lidar, and radar, from various driving scenarios.

• Feature Extraction: Extract features using CNNs and RNNs for image and sequential data, respectively.

• Model Architecture: Design a suitable ANN architecture with convolutional, recurrent, and fully connected layers.

• Training and Tuning: Train the model on a diverse dataset, optimize hyperparameters, and validate performance regularly.

• Safety Measures: Implement emergency braking and collision avoidance mechanisms to prioritize safety.

• Deployment: Deploy the system initially in controlled environments, expanding gradually to more complex scenarios.

• Continuous Improvement: Monitor real-world performance, collect new data, and continuously improve the model's decision-making.